HIMANSHU SINGH

Bareilly, U.P. | singhhimanshu33456@gmail.com | +91 8077005643
linkedin.com/in/himanshu-singh07/ | github.com/Himanshusinghdev1

SUMMARY

Computer Science student specializing in AI/ML engineering, proficient in Python and deep learning with hands-on experience in TensorFlow, PyTorch, and Keras. Built end-to-end applications including RAG chatbots and CNN classifiers achieving 96%+ accuracy in academic projects. Seeking an entry-level AI/ML engineer role to contribute software development, model deployment, and modern AI framework expertise.

EDUCATION

Bachelor of Technology in Computer Science & Engineering **2026**
Invertis University, Bareilly | Percentage: 79%
Relevant Coursework: Machine Learning, Deep Learning, NLP, DSA

Senior Secondary (Class 12), CBSE **2022**
Sobtis Public School, Bareilly | Percentage: 67%
Stream: Science (Physics, Chemistry, Mathematics, Computer Science)

SKILLS

Languages: Python, SQL, Java
Data Science: Machine learning, Deep Learning, Natural Language Processing, Generative AI,
Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, SpaCy, Streamlit, Flask, Langchain
Tools: GitHub, DagsHub, FAISS, MS Excel, MLflow, Groq API, VS Code
Platforms: AWS, GCP, Docker, Jupyter Notebook, Kaggle, Hugging Face, Together.ai

PROJECTS

AskScout - LLM-Powered RAG Chatbot | **(**RAG, LLaMA 4, Groq & Cohere Embeddings**) | LINK** **2025**

- Developed a **RAG-based Streamlit chatbot** leveraging **Meta LLaMA 4 Scout via Groq API** for intelligent, context-aware question-answering.
- Integrated Cohere's English-V3 embeddings with **FAISS** to power semantic search across 100+ user-supplied URLs, improving retrieval accuracy by 40%.
- Deployed a scalable app with real-time content extraction via LangChain, reduced response latency by 60% using Groq's high-speed LPU acceleration.

Automatic Number Plate Recognition | **(**YOLOv5, EasyOCR, Python, OpenCV, MLOps**) | LINK** **2025**

- Developed ANPR system using **YOLOv5** and **EasyOCR** achieving **99.5% mAP** detection accuracy with real-time video processing at **2-8 FPS**.
- Built complete **MLOps** pipeline with **7 automated stages** including data preprocessing, model training, and validation, reducing processing time by 80%.
- Implemented computer vision solution handling 10,000+ annotated images with XML-to-YOLO conversion and OCR text recognition across **multiple formats**.

3-class Lung Cancer Classification Web App | **(**VGG16, keras, Flask, DagsHub, AWS**) | LINK** **2024**

- Developed a deep learning model for automated lung cancer classification using histological microscopic tissue images, enabling the research team to identify **top 3 drivers** of tumor growth.
- Implemented **CNN** architecture with **VGG16** transfer learning, achieving **96.09% accuracy** and **0.1112 loss** on validation dataset.
- Constructed a scalable image processing pipeline with **Flask**, **DagsHub MLOps**, and **AWS**, reducing image processing time by **30%** and improving the speed of lung cancer diagnosis.

CERTIFICATION

- *Generative AI Professional Certificate* [**Google Cloud Platform**] | **CERTIFICATE** **Apr-Jun 2025**
- *Deep Learning **(ELITE)*** [**NPTEL, IIT Kharagpur**] | **CERTIFICATE** **Jan-Apr 2025**
- *Introduction to Machine Learning **(ELITE)*** [**NPTEL, IIT Kharagpur**] | **CERTIFICATE** **Jul-Sep 2024**